SIH 1505 — INDUSTRIAL EMERGENCY PRE-PLAN

INCIDENT ID: INC-T-01-14K
DOCUMENT VERSION: v1.0 [AUTHORIZED (PROTOTYPE DEMO)]

FACILITY: PetroChem Complex Alpha - Unit 04
PROTOTYPE FACILITY REF: PCH-ALPHA-04 (Demo Facility — Non-Statutory Evaluation)

GENERATED: 2026-08-15 17:27:13 UTC
PREPARED BY: SIH-1505 Decision Support Engine

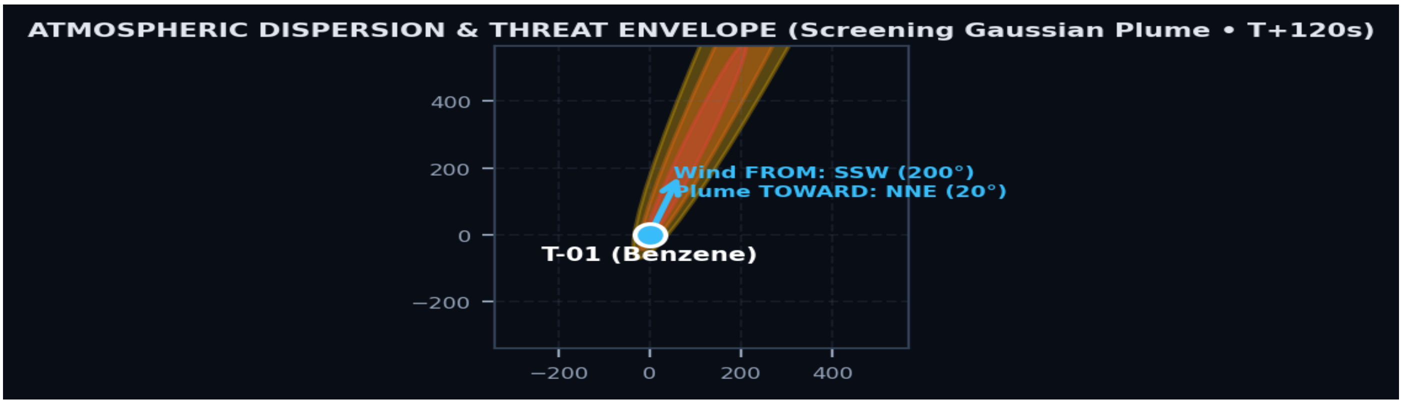
INCIDENT SEVERITY: MODERATE (SCORE: 38.6/100) • GOVERNANCE: AUTHORIZED BY DR. KAVITA KRISHNAN (Chief Safety Officer (HSE))

1. INCIDENT SOURCE, CHEMICAL SDS & METEOROLOGICAL PROFILE

Incident Type:	Tank Leak	**Source Asset:**	T-01 (21.6865, 72.5710)
Hazardous Substance:	**Benzene (Pure Grade)**	**Release Emission Rate:**	20.0 kg/s (Duration: 30 min)
Meteorological Wind Vector:	14.5 km/h • **FROM:** SSW (200°)	**Plume Propagation Vector:**	Downwind Travel • **TOWARD:** NNE (20°)
Ambient Temperature:	**29.0°C**	**Weather Telemetry Source:**	DEMO (Scenario Preset) • Stability: Class D

2. HAZARD THREAT ZONES (SCREENING GAUSSIAN DISPERSION ENVELOPE)

Zone Tier	Threshold Criterion	Downwind Reach	Crosswind Width	Threat Area
Red Threat Zone	ERPG-3 / High Lethality (1000.0 ppm)	664.9 m	47.4 m	20,621 m²
Orange Threat Zone	ERPG-2 / Irreversible Injury (150.0 ppm)	1930.4 m	123.6 m	155,779 m²
Yellow Threat Zone	ERPG-1 / Mild Irritation (50.0 ppm)	3000.0 m	215.5 m	445,394 m²



3. POPULATION & INFRASTRUCTURE EXPOSURE IMPACT

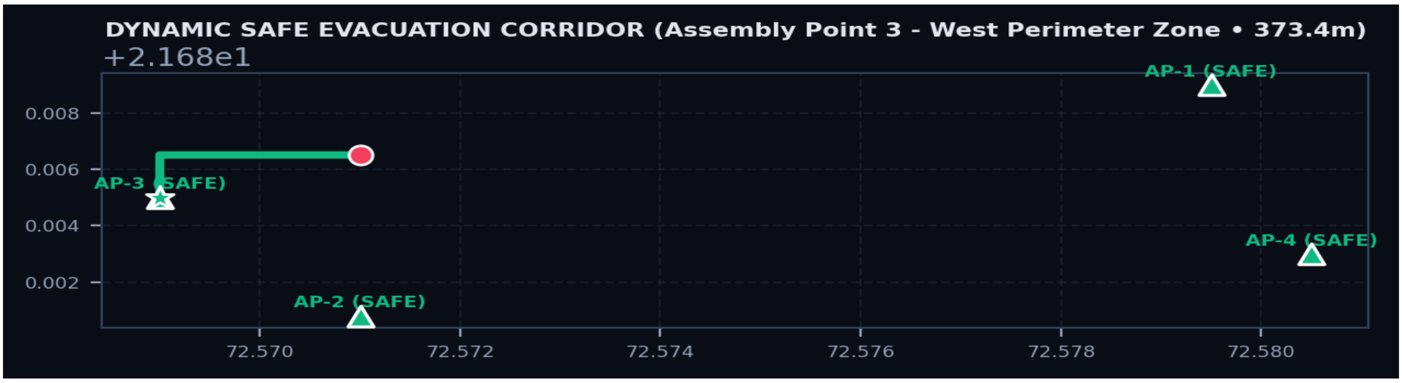
Total Workers on Site:	28 Personnel **Exposure Assessment:** No active seeded worker coordinates intersected the calculated threat envelopes at simulation time.	**Lethal Zone (Red) Exposed:**	**0 Workers**
Severe Zone (Orange) Exposed:	0 Workers	**Caution Zone (Yellow) Exposed:**	0 Workers
Compromised Plant Assets:	1 of 16 Units	**Severed Road Segments:**	4 of 10 Segments

4. DYNAMIC SAFE EVACUATION CORRIDOR & MUSTER DIRECTIVES

Designated Safe Assembly Point:	**Assembly Point 3 - West Perimeter Zone**	**Exit Perimeter Gate:**	Gate 2 - South Commercial & Tanker Logistics Gate
Total Safe Egress Distance:	373.4 meters	**Estimated Walk Time:**	5.2 minutes (at 1.2 m/s)
Multi-Factor Scoring:	Safety: 70% | Dist: 76% | Rank: **0.727**	**Route Clearance Status:**	**DIVERTED**
Selection Rationale:	Selected as optimal Upwind/Crosswind egress corridor (149° clearance from plume axis) providing shortest uncompromised route to Assembly Point 3 - West Perimeter Zone (Safety Score: 0.70, Composite Score: 0.73).	**Avoided Blocked Roads:**	Perimeter Ring Road - West, Perimeter Ring Road - South, Chemical Sector Cross Road (East-West North), Process Area Cross Road (East-West South)

Evaluated Muster Point Alternatives & Rejection Analysis:

Candidate Assembly Point	Exit Gate	Distance	Safety	Status / Rejection Reason
ROUTE-AP-3 (AP-3)	GATE-2	373.4 m	70%	**SELECTED:** None
ROUTE-AP-2 (AP-2)	GATE-2	1047.1 m	70%	**VIABLE_BACKUP:** Internal access roads severed; requires outer peripheral detour
ROUTE-AP-1 (AP-1)	GATE-1	1569.5 m	70%	**REJECTED:** Internal access roads severed; requires outer peripheral detour
ROUTE-AP-4 (AP-4)	GATE-3	1784.1 m	70%	**REJECTED:** Internal access roads severed; requires outer peripheral detour



5. TACTICAL EMERGENCY RESOURCE DEPLOYMENT & SUPPRESSION STRATEGY

DECISION-SUPPORT RECOMMENDATION — REQUIRES SITE/HSE VALIDATION: All tactical quantities, PPE ensembles, suppression demands, and response actions are prototype computational recommendations and must be validated against facility ERDMP and competent safety authorities before operational deployment.

Upwind Staging Standoff:	250.0 meters	**Isolation Cordon Radius:**	831.1 meters
Firewater Demand:	6,700 LPM	**Foam Concentrate Demand:**	2,500 Liters (AFFF 3%)
Mandatory Entry PPE:	**Level A Fully Encapsulated Gas-Tight Suit (Trellchem) with positive-pressure SCBA (NIOSH Certified)**	**Calculation Basis:**	Calculated from release rate 20.0 kg/s, 1 vulnerable assets, and Benzene (Pure Grade) SDS toxicity/flammability

Unit Name & Type	Station	Staging Geolocation	Transit	ETA	Priority
Fire Tender Alpha (10,000L Multi-Purpose Water/Foam) Vapor Cloud Inerting & Preemptive Foam Blanket	FS-01	Upwind Incident Staging Post (Bearing 20°, 250m Standoff)	736.2 m	**2.5m**	**IMMEDIATE**
Hazmat Rapid Response Unit (Level A/B Ensembles) Hot Zone Isolation & Emergency Valve Clamping	FS-01	Incident Hot-Zone Control Post (20° Upwind)	726.4 m	**2.5m**	**IMMEDIATE**
Industrial Emergency Response Team (ERT Alpha) Cordon Enforcement, Road Junction Barricading & Headcount Control	CR-01	Main Outer Access Gate & Security Command	303.6 m	**1.3m**	**SUPPORT**
High-Volume Water Curtain Bowser Secondary Water Supply & Runoff Dilution	FS-01	Crosswind Knockdown Sector (110° Vector)	463.8 m	**1.8m**	**SUPPORT**

Advanced Cardiac & Toxic Resuscitation Ambulance Precautionary Medical Standby at Muster Point	MED-01	Medical Triage at Assembly Point 3 - West Perimeter Zone	805.5 m	**2.7m**	**STANDBY**

6. INCIDENT COMMAND EXECUTIVE SUMMARY SNAPSHOT

Risk:	38.6/100 — MODERATE	**Chemical:**	Benzene (Pure Grade)
Personnel Exposed:	0 workers (Red: 0)	**Compromised Assets:**	1 of 16 units
Blocked Roads:	4 segments	**Primary Muster:**	**Assembly Point 3 - West Perimeter Zone**
Exit Gate:	Gate 2 - South Commercial & Tanker Logistics Gate	**Egress Metrics:**	373.4 m / 5.2 min
Lead Tactical Resource:	Fire Tender Alpha (10,000L Multi-Purpose Water/Foam) (IMMEDIATE, ETA: 2.5m)	**Firewater Demand:**	6,700 LPM Water (2,500 L AFFF Foam)

7. STANDARD OPERATING PROCEDURE (SOP) PHASED ACTION DIRECTIVES

Phase 1: Initial Alarm, Isolation & Personnel Evacuation (0 - 5 Minutes)

• Sound Plant Emergency Siren (Continuous Alarm for Benzene (Pure Grade) Tank Leak).

• Actuate Emergency Shutdown (ESD) for T-01 and isolate upstream pipeline valves.

• Enforce mandatory crosswind evacuation of all Sector personnel towards Assembly Point 3 - West Perimeter Zone.

• ERT Team establishes 831.1m initial cordon perimeter around T-01.

Phase 2: Source Mitigation & Tactical Exposure Protection (5 - 15 Minutes)

• Position Fire Tender & Bowser at Upwind Staging Post (250m standoff, bearing 20°).

• Establish 6,700 LPM water curtain coverage to suppress Benzene (Pure Grade) vapors and cool 1 adjacent assets.

• Deploy Hazmat Entry Squad in Level A Fully Encapsulated Gas-Tight Suit for hot-zone valve isolation and clamp seal.

• Verify automated fixed water deluge systems on neighboring storage vessels.

Phase 3: Containment, Medical Triage & Incident De-escalation (15 - 30 Minutes)

• Complete headcount verification at Assembly Point 3 - West Perimeter Zone (Reconcile against 0 exposed workers).

• Ambulance team initiates emergency triage and decontamination for exposed operators.

• Contain all contaminated firefighting water in industrial effluent retention sump; prevent municipal drain escape.

• HSE Lead Commander compiles situation briefing for District Disaster Management Authority (DDMA).

8. DATA TRACEABILITY & SCREENING MODEL ASSUMPTIONS

Dispersion Physics:	Screening Gaussian Approximation with Pasquill-Gifford stability parameters	**Human Governance:**	Mandatory HSE human review and authorization required prior to operational execution
Weather Telemetry:	DEMO Telemetry (Scenario Preset) • Recorded at 29.0°C, 14.5 km/h	**Statutory Scope:**	Decision-support prototype only; non-certified under statutory OISD/ALOHA rules
Terrain Modeling:	Simplified 2D/3D flat terrain representation with standard surface roughness (z■ = 0.5m)	**Road & Worker Graph:**	Prototype facility CAD/GIS network with obstacle avoidance pathfinding

9. REGULATORY DISCLAIMER & PROTOTYPE NOTICE

PROTOTYPE DECISION SUPPORT — NON-CERTIFIED COMPUTATIONAL WORKFLOW:
This emergency pre-plan document is prepared automatically by the SIH 1505 Decision Support Engine. Dispersion envelopes, routing corridors, and tactical dispatches are computed for hackathon demonstration and decision-support prototyping only. This software does NOT claim certified ALOHA equivalence or statutory compliance. Operational tactical actions require validation against the facility's approved Emergency Response and Disaster Management Plan (ERDMP), applicable OISD/PESO standards, and competent statutory safety authorities.

10. HUMAN-IN-THE-LOOP PRE-PLAN AUTHORIZATION & ENDORSEMENT

DOCUMENT STATUS: AUTHORIZED (PROTOTYPE DEMO)
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Demonstration Approver: Dr. Kavita Krishnan
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Approval Notes: Benzene Vapor Containment Authorized.

DEMONSTRATION SIGNATURE BLOCK:
✓ AUTHORIZED (PROTOTYPE DEMO)
Dr. Kavita Krishnan
Chief Safety Officer (HSE)
DEMO SIGNATURE — NOT A REAL SIGNATURE
(Ready for PKI Digital Signature Integration)